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Coordinated Optimization of Emergency Supply Distribution and Multi-Type Personnel Evacuation under Hazardous-Material Vehicle Incidents on Urban Freeways

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Abstract. Hazardous-material incidents on urban freeways trigger compound disasters requiring coordinated supply distribution and evacuation. Using a hypothetical chlorine release on Beijing's Sixth Ring Road, this study develops an integrated framework that combines Gaussian-plume ERPG zoning with co-optimization of a truck-drone logistics network and a five-mode evacuation fleet across three vulnerability classes. ERPG-based zoning reduces the active evacuation population from 235,000 to 13,800, while shelter-in-place demand raises supply tasks from 40 to 70. Under lightly damaged roads, the truck-only scheme outperforms the truck-drone scheme, exposing the applicability boundary of relay delivery. Vulnerable groups achieve 100% evacuation, but critically injured victims reach only 2.8%, revealing a structural medical-fleet bottleneck. The framework uncovers an asymmetric "high-supply / low-evacuation" resilience pattern that informs urban emergency decisions.

Keywords: Emergency logistics, truck-drone coordination, multi-type evacuation, hazardous-material incidents, urban expressways, transportation resilience.

1 Introduction

Urban freeways carry both commuter and freight flows and serve as principal corridors for hazardous-material (hazmat) transportation. When an incident occurs, consequences such as toxic gas release or explosion can trigger compound disasters that require coordinated multi-agency response[1]. Recent accidents in China—e.g., the 2005 Huai'an liquid-chlorine rollover (29 deaths, ~10,000 evacuated) and the 2020 Wenling tanker explosion (20 deaths)—expose the vulnerability of urban transport networks. The Beijing Sixth Ring Road, with a total length of about 188 km, has experienced multiple hazmat-related incidents in recent years (involving diesel, liquefied petroleum gas, etc.); its southeastern Taihe Bridge segment is densely populated and concentrated with critical infrastructure, making it a high-risk node. How to achieve the coordinated optimization of emergency supply distribution and personnel evacuation under stringent time constraints is a critical scientific question that urgently needs to be addressed.

Emergency logistics has been extensively reviewed[1, 2], with recent work addressing demand uncertainty and link disruption[3]. Truck-drone coordination, originating from the flying-sidekick TSP[4, 5] and surveyed in[6, 7], has been extended to disaster relief with priority mechanisms[8]. Evacuation modeling has evolved from network-flow formulations[9] to multimodal frameworks[10] and dynamic multi-vehicle dispatch[11], complemented by domestic studies on urban rail evacuation[12], network resilience[13], and cooperative control[14]. For hazard-zone evaluation, the Gaussian plume model combined with the Pasquill-Gifford stability classification and the ERPG threshold system remains the standard practical framework.

However, most existing studies treat supply distribution and personnel evacuation as independent problems[1, 3], overlooking their interdependence in resource competition, temporal coupling, and spatial interaction. Taking a chlorine release at the Taihe Bridge area on the Beijing Sixth Ring Road as the scenario, this study develops a coordinated framework that simultaneously optimizes emergency supply distribution and multi-type personnel evacuation. The contributions are: (1) joint optimization of supply distribution (truck-drone coordination, four priorities, time windows) and multi-type evacuation (five modes, three vulnerability classes) under a unified scenario; (2) a high-fidelity simulation environment built on multi-source urban data (mobile-phone OD, POI, census-based age structure, OSM); (3) integrated scenario analyses covering Gaussian plume, wind-direction sensitivity, and resilience assessment; and (4) practical guidance on pre-positioning, fleet sizing, and multimodal coordination.

2 Study Area and Data

2.1 Study area

The Taihe Bridge Interchange (116.494°E, 39.736°N) on the southeastern Sixth Ring

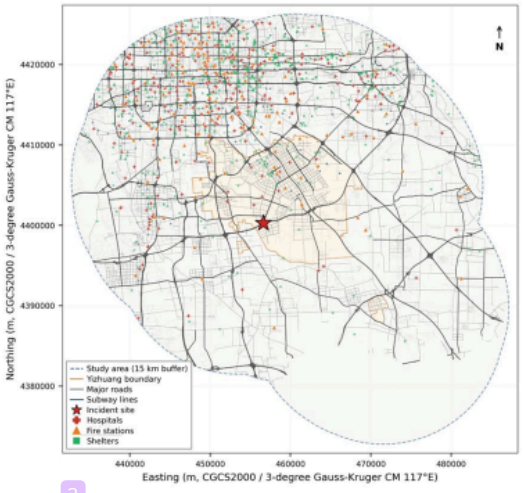


Fig. 1. Overview of the study area showing the road network and emergency facilities.

Road is selected as the incident epicenter. We define a 15-km buffer area covering approximately 707 km² to ensure full coverage of potential toxic gas impacts and surrounding emergency facilities. Figure 1 shows the spatial extent of the study area, the road network, and the distribution of emergency facilities.

2.2 Data sources

This study integrates seven categories of multi-source urban data to construct a high-fidelity simulation environment. The sources, temporal coverage, spatial precision, and key attributes of each dataset are summarized in Table 1.

Table 1. Summary of data sources.

Data category	Source	Spatial precision	Key attributes
Road network	OpenStreetMap	Segment level	Functional class, speed limit, direction
Points of interest	Amap (Gaode)	Point	Sub-category, name, WGS84 coordinates
Population spatiotemporal distribution	Baidu mobile-phone positioning	100 m grid	Hourly OD flows, commuting modes
Block-level population	Baidu LBS	Block	Resident population count
Age structure	Seventh National Population Census (2020)	District/county	Age-sex distribution by 5-year cohort
Commuting patterns	Baidu mobile-phone positioning	100 m grid	Modal split (car, metro, bus, etc.)
Administrative boundaries	Planning authority	Polygon	Yizhuang Development Area regulatory boundary

2.3 Research framework

We adopt a three-layer "scenario-driven, dual-chain coordination, resilience assessment" framework. Layer 1 uses the Gaussian plume model to simulate chlorine dispersion and generate a damaged road network with affected population as shared inputs. Layer 2 couples two optimization chains on this network: a "centripetal" supply chain (truck-drone logistics minimizing makespan and time-window penalties) and a "centrifugal" evacuation chain (multimodal dispatch minimizing weighted evacuation time and risk exposure). Layer 3 assesses resilience along three dimensions with explicit indicators: structural resilience (closure length and ratio within the influence zone), functional resilience (supply-task completion rate, on-time life-rescue rate, evacuation completion rate by vulnerability class, and cumulative risk exposure), and adaptive resilience (marginal improvement from technology- or policy-driven interventions, such as truck-drone relay or fleet expansion).

Fig. 2. Technical roadmap of the coordinated emergency response framework: scenario-driven, dual-chain coordination, and resilience assessment.

3 Methodology

3.1 Scenario modelling

Chlorine atmospheric dispersion is simulated using the Gaussian plume model. The scenario assumes a 5 kg/min release rate with a 3.0 m/s southwest wind. Hazard zones are defined by ERPG thresholds: ERPG-3 indicates a core hazard zone requiring mandatory evacuation. ERPG-2 is a warning zone recommending urgent evacuation. ERPG-1 is an influence zone recommending shelter-in-place. We also incorporate BPR-based traffic congestion effects and a 1 km core road-closure zone.

3.2 Emergency supply distribution model

We construct a three-tier logistics network comprising supply points, transshipment hubs, and demand points. Supplies are classified into four priority categories, including life-rescue supplies with strict 30-min hard time windows. A two-stage algorithm (greedy construction and 2-opt local search) evaluates both direct truck and relay (truck-drone) deliveries; relay is triggered when the destination lies inside the closure zone or the direct truck travel time exceeds a threshold. The objective minimizes a weighted sum of makespan, time-window penalties, and unmet demand, with the unmet-demand penalty applied per missed task and weighted by priority class. Key model parameters - the 5 kg/min release rate, the fleet (15 ambulances, 3 helicopters, plus buses, private cars, metro trains), and supply-point inventory - are calibrated to historical tanker-incident magnitudes and the actual emergency-resource inventory of the study area; the 2% k=1 share in ERPG-1 is a conservative upper bound used to stress-test medical capacity.

3.3 Multi-type evacuation model

A risk-aware time-stepping dispatch model coordinates five transport modes: ambulances, helicopters, buses, private cars, and metros. Dispatch occurs in 5-min steps over a 180-min horizon. The algorithm prioritizes evacuees dynamically based on vulnerability class (k=1 first) and real-time risk exposure. It minimizes weighted evacuation time, dynamic risk exposure, and unmet-demand penalties.

4 Case Study and Numerical Results

4.1 Incident scenario configuration

Within the 1 km core-closure zone, 96.7 km of road is closed, accounting for 19.1% of the total road length (505.3 km) within the 5 km influence zone. Based on ERPG concentration contours (after buffering), three response zones are delineated: the ERPG-2 warning zone (≥3 mg/m³ buffer) contains approximately 19,700 people, and the ERPG-1 influence zone (≥1 mg/m³ buffer) contains approximately 407,000 people. Under the differentiated zoning strategy, approximately 13,800 people require active evacuation (k=1 and k=2 in the ERPG-2 zone, plus k=1 in the ERPG-1 zone), while approximately 413,000 people shelter in place (generating supply-distribution demand).

4.2 Wind-direction sensitivity analysis

Four wind scenarios (225° baseline; 0°; 90°; and 225° at 8 m/s) confirm that wind direction decisively shifts which population centers and infrastructure clusters fall within hazard zones, underscoring the need for real-time meteorological integration.

4.3 Supply distribution results

Table 2 compares the supply-distribution schemes. Shelter-in-place demand from ~413,000 people increases the workload from 40 to 70 tasks, evidencing the substantive coupling between evacuation and distribution. The truck-only scheme completes 61 tasks (87.1%), with all life-rescue supplies delivered within the 30-min hard time window. The truck-drone scheme expands relay capacity (4 relay tasks vs. 0 in the truck-only baseline), effectively covering demand points inside the closure zone, but completes 1 fewer task and its makespan rises from 47.3 to 59.9 min (+26.6%) because relay consumes vehicle time. The drone-only scheme is infeasible since most supplies exceed the 25 kg drone payload limit. The 6 unfinished tasks concentrate on bulk protection-equipment delivery, reflecting supply-point inventory constraints that could be alleviated by adding emergency warehouses.

Table 2. Supply delivery scheme comparison.

Scheme	Completed	Makespan	Mean time (min)	Penalty	Direct	Relay	Improvement
Truck-only	61	47.3	24.2	0	61	0	—
Truck–UAV	60	47.3	25.6	0	56	4	0%
UAV-only	0	—	—	—	0	0	—
Cooperative (doubled UAVs)	60	47.3	25.6	0	56	4	0%

4.4 Evacuation results

Figure 3 compares evacuation schemes. The ERPG zoning strategy reduces the actively evacuated population from 235,000 (traditional circular range) to 13,800; the baseline achieves 41.0% within 180 min (5,671 evacuees), substantially above the 9.1% without zoning. All 5,302 k=2 vulnerable evacuees in the ERPG-2 zone are fully evacuated

(100%), and the overall ERPG-2 rate reaches 93.1%, validating priority-based scheduling. However, k=1 critically injured victims become a new bottleneck: limited to ambulances and helicopters, the estimated 8,100 k=1 population in the ERPG-1 zone (at 2%) far exceeds the capacity of 15 ambulances and 3 helicopters, yielding only a 2.8% rate (243 evacuees). Sensitivity analyses show that removing helicopters lowers the rate to 40.2% and raises cumulative risk by 6.4% (85,694 → 91,153), while removing the metro lowers it to 38.8%, confirming the metro's meaningful contribution to k=2/k=3 capacity. Flexible and strict priority schemes coincide at 41.0% because no k=3 demand is active under the current scenario.

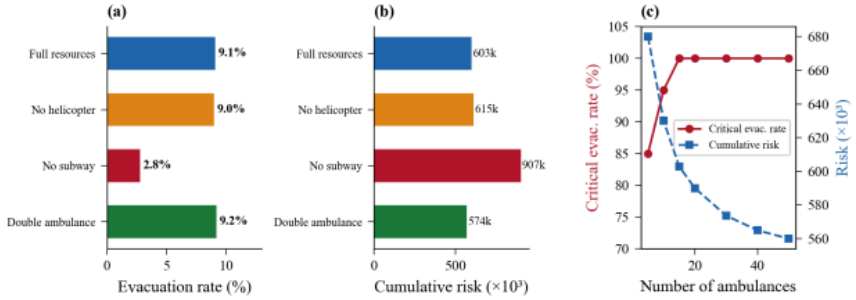


Fig. 3. Evacuation optimization results: (a) Comparison of evacuation completion rates; (b) Comparison of cumulative risks; (c) Ambulance sensitivity.

5 Discussion

5.1 Asymmetric resilience between supply distribution and evacuation

The system shows a pronounced asymmetric resilience pattern: supply distribution achieves high functional resilience, while k=1 critical-casualty evacuation forms a sharp structural bottleneck. The root cause is the mismatch between dedicated medical capacity (18 ambulances and helicopters) and k=1 demand (~8,100 in the ERPG-1 zone under a 2% rate), which is orders of magnitude larger than the fleet can absorb. Shelter-in-place demand further constrains the supply side, where 6 unfinished bulk-protection-equipment tasks reflect supply-point inventory limits. These findings indicate that pre-positioning of dedicated emergency-medical assets and emergency-warehouse layout deserve top priority in urban hazmat preparedness.

5.2 The decisive role of critical infrastructure

The metro's evacuation contribution is substantial but easily underestimated: removing it lowers the rate by 2.2 percentage points (41.0% → 38.8%), confirming its supplementary role for k=2/k=3 even when k=1 and k=2 dominate. Its value grows under larger-scale incidents (e.g., ERPG-3 covering dense residential clusters) where k=3 throughput becomes critical, and shelter-in-place residents tend to spontaneously gather at stations where high-capacity service (1,200 passengers per train, 15 trains per 180 min) is irreplaceable. Consistent with [10], this calls for cities to establish rapid emergency-dispatch protocols ensuring prompt metro activation after a hazmat incident.

5.3 Applicability boundary of truck-drone coordination

Shelter-in-place demand raises relay deliveries from 0 to 4 tasks, yet the truck-drone scheme yields a counterintuitive negative result: makespan rises 26.6% (47.3 → 59.9 min) and 1 fewer task is completed. Three causes are identified: (i) stricter POI filtering reduces fire stations from 1,032 to 331, tightening the supply-point pool; (ii) the road-closure ratio is only 19.1%, leaving direct routes feasible; and (iii) the relay-trigger threshold is too low, mis-assigning tasks to relay. This is consistent with [8] that drones add value mainly under large-scale damage. In lightly damaged scenarios, schedulers should prioritize direct delivery and trigger relay only when direct travel time exceeds a threshold or the destination lies inside the closure zone; advantages of relay grow as damage intensifies.

6 Conclusions

6.1 Main findings

This study develops a coordinated framework for emergency supply distribution and evacuation, applied to a hypothetical chlorine release at Beijing's Taihe Bridge. Three findings stand out: (1) ERPG-based zoning cuts evacuees from 235k to 13.8k while raising supply tasks 75% (40→70), forming a "centripetal-centrifugal" coupling. (2) The system shows asymmetric resilience—87.1% supply completion and 100% life rescue, but only 2.8% k=1 evacuation—due to a mismatch between medical capacity and critical-casualty demand. (3) Truck-drone relay is not universally beneficial: under lightly damaged networks (19.1% closure), it increases makespan by 26.6%. These conclusions are scenario-specific; future work should refine injury models and conduct multi-scenario sensitivity analysis.

6.2 Policy recommendations

(1) Develop and refine emergency-operation protocols for the metro system, ensuring rapid activation of emergency service after a hazmat incident. (2) Adopt layered evacuation: mandatory evacuation within the 1 km core zone, shelter-in-place with protective-equipment distribution in the 1-3 km outer zone. (3) Pre-position helicopter assets along high-risk corridors to enable rapid transport of critical casualties. (4) Invest in drone infrastructure: under more severe damage conditions, the advantages of relay delivery will become more pronounced. (5) Integrate real-time wind monitoring with evacuation planning to dynamically update hazard zones and trigger zone-based evacuation orders based on meteorological data.

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